

1.

1(a) Expression: $B'C + B'C'D$

$$B'(C + D)$$

1(b) Expression $(Y' + Z')(Y + Z')$

$$YZ'$$

1(c) Expression $C'D'A + CD'A + DA$

$$A$$

1(d) Expression: $X(C'Y + C'Y') + X(W'C' + W'C)$

$$X(C' + W')$$

1(e) Expression: $(B'C' + BD + C'D)(B' + C + B'C)$

$$B'C' + BCD$$

2.

2(a) Expression $Y + Z'(W + Y' + W')$ (to two literals)

$$Y + Z'$$

2(b) Expression $Y'X + YX'Z + Y'X'$ (to three literals)

$$Y' + X'Z$$

2(c) Expression $Y'X(W' + ZW) + X(Y + Y'Z'W)$ (to one literal)

X

3.

3(a) Only OR and complement

$$(X + Z')' + (X' + Z + Y')' + (X + Y)'$$

3(b) Only AND and complement

$$((X'Z)'(XZ'Y)'(X'Y')')'$$

4.

4(a) Expression: $B'D' + BD$

$$(B + D)(B' + D')$$

4(b) Expression $(C + B'D)(C + B' + D')(C'B' + D')$

$$(D + C')(B + C + D')$$

4(c) Expression: $W'Y'(ZX' + Z'X) + WY(Z + X)(Z' + X')$

$$XZ + WY' + W'Y + X'Z'$$

4(d) Expression: $(B' + D')AC + E$

$$(BD + A' + C')E'$$

5.

5(a) Expression: $(A' + B'D')(D' + AC')$

SOP $A'D' + B'D'$

POS $D'(A' + B')$

5(b) Expression : $(Z' + Y)X'(X' + Z) + X$

SOP $X + Y + Z'$

POS $(X + Y + Z')$

5(c) Expression: $(Z' + XW + WY')(X' + Z'Y')$

$X'Z' + X'W'Y + XWZ' + W'YZ'$

$(X' + Z')(X + Y + Z')(W' + X')(W' + Z')$

6.

6(a) $B + D$

6(b) $XY' + X'Z'$

6(c) $Z' + WX$

7.

7(a)

$$WXZ' + WZY' + XZW' + W'Y'Z'$$

7(b)

$$AD + A'C' + B'D'$$

7(c)

$$AC + CB' + B'D' + A'BD$$

7(d)

$$XZ + YZ' + X'Y'$$

7(e))

$$WZX' + WY'Z' + W'Y'Z + W'X'Z'$$

7(f)

$$BC + ADC' + A'B'D'$$

8.

8(a)

$$W'$$

8(b)

$$BC + ACD + A'B'D'$$

8(c)

$$BC + CD' + ADB'C'$$